



## Research papers

# A Bayesian framework to assess and create risk maps of groundwater flooding

Pablo Merchán-Rivera<sup>a</sup>, Alexandra Geist<sup>a</sup>, Markus Disse<sup>a</sup>, Jingshui Huang<sup>a</sup>, Gabriele Chiogna<sup>a, b, \*</sup>

<sup>a</sup> Chair of Hydrology and River Basin Management, Technical University Munich, Arcisstrasse 21, 80333 Munich, Germany

<sup>b</sup> Institute of Geography, University of Innsbruck, Innrain 52, 6020 Innsbruck, Austria

## ARTICLE INFO

This manuscript was handled by Corrado Corradini, Editor-in-Chief, with the assistance of Saket Pande, Associate Editor

## Keywords:

Groundwater flooding  
Bayesian inversion  
Probability maps  
Flood risk  
Sensitivity analysis  
Uncertainty quantification

## ABSTRACT

Groundwater flooding can cause severe damages to homes, utilities, and infrastructure and yield significant economic and social costs. Numerical models are used to understand these events and are the basis to produce imagery products for risk management and communication. However, such maps are generally produced using forward model simulations, and most of the mapping products are still deterministic. In contrast to pluvial and fluvial floods, an open issue in the analysis of susceptibility to groundwater flooding is the lack of probabilistic assessment and mapping products recognizing parametric uncertainty. Hence, we propose a Bayesian-based framework to create probabilistic risk maps and to identify the susceptibility to groundwater flood events. We aim to assess the spatial distribution and temporal variability of groundwater flooding by decomposing the uncertainty and the sensitivity of distributed groundwater numerical models. The scheme involves the use of the elementary effects method, the DiffereNTial Evolution Adaptive Metropolis (DREAM) algorithm, and the exploration of the predictive posterior distributions of the groundwater heads to evaluate the susceptibility according to exceedance levels. We use the proposed Bayesian framework with a numerical model that simulates the groundwater flood event that occurred in the valley of the Alz River in 2013. This study developed two types of susceptibility maps based on the exceedance probability of certain groundwater levels and specific cellar depths. The Bayesian inference supports the parameter estimation and thereby increases the spatial confidence of the areas susceptible to inundation. This study shows that maps of susceptibility to groundwater flooding can be built over one single event models and acknowledge the inherent spatial and temporal uncertainty of such events.

## 1. Introduction

The increasingly frequent occurrence of extreme weather events has raised attention about flood events (Kaiser et al., 2020; Tabari, 2020; Winsemius et al., 2016). Although the focus of flood risk assessment often tends to be on the fluvial and pluvial floods (Ascott et al., 2017; Collins et al., 2020), groundwater flooding can yield significant economic and social damages and disruptions, such as sewer backups (Abboud et al., 2018; Morris et al., 2018), roads and property inundation (Abboud et al., 2018; Keilholz et al., 2015; Morris et al., 2018; Soren, 1976), foundation failures and corrosion (Colombo et al., 2018; Oyedele et al., 2009), damages to underground structures and infrastructure (Gattinoni and Scesi, 2017; Preene and Fisher, 2015), underground pollution transport (González-Quirós and Fernández-Álvarez, 2019), and crop damage (Booth et al., 2016). This type of flood event refers to the groundwater emergence at the ground surface and the rising groundwater level into underground civil infrastructure

(Macdonald et al., 2012; Macdonald et al., 2008). Unlike other types of floods, groundwater flooding can be difficult to identify and assess (Kreibich et al., 2009; McKenzie et al., 2010), and traditional flood protection systems may not be effective (Yu et al., 2019).

Similar to traditional flood risk management, a large number of studies applies numerical modeling at different scales and with a varying degree of complexity to understand groundwater flood events (Abboud et al., 2018; González-Quirós and Fernández-Álvarez, 2019; Morrissey et al., 2020; Yu et al., 2019). These models are typically posed as forward problems, which involve predicting “error-free” model states based on a prior parameterization of the model (Tarantola, 2005). One disadvantage in this type of strategy is that it requires specifying a variety of parameters, which are impossible to characterize exhaustively (Zhou et al., 2014). Another drawback of forward problems is that, once calibrated, the model ignores the equifinality thesis (Beven and Freer, 2001) and the uncertainties, which may imply to adopt strong and questionable assumptions about its forecast reliability (Di

\* gabriele.chiogna@tum.de

E-mail address: [gabriele.chiogna@tum.de](mailto:gabriele.chiogna@tum.de) (G. Chiogna).

<https://doi.org/10.1016/j.jhydrol.2022.127797>

Received 23 November 2021; Received in revised form 1 March 2022; Accepted 3 April 2022  
0022-1694/© 20XX

Baldassarre et al., 2010). Consequently, probability theory tools, such as inverse methods (Helton et al., 2004; Kaipio and Somersalo, 2005), have evolved to address aleatory and epistemic uncertainties (Merz and Thielen, 2009; Ross et al., 2009). In contrast to forward models, inverse methods were developed to support parameter identification and consequently to improve predictions (Zhou et al., 2014). Bayesian approaches solve inverse problems considering parameters as random variables and providing posterior probabilities by incorporating known data (Smith, 2013). Due to their inferential and predictive properties, Bayesian methods are considered one of the most appropriate techniques for uncertainty analysis (Heße et al., 2019). In spite of the progress in the field of uncertainty quantification, the temporal and spatial uncertainties of the models are often declared but not quantified in groundwater flooding studies. A limited number of studies includes a formal quantification of the uncertainty (Colombo et al., 2018; Fürst et al., 2015; Jimenez-Martinez et al., 2015), and, to the best of the authors' knowledge, the explicit application of Bayesian inference on groundwater flooding modeling remains unexplored.

Flood risk mapping is a fundamental tool for flood risk management and communication (Demeritt and Nobert, 2014; Henstra et al., 2019). Accordingly, a lot of effort has been made to generate groundwater flood maps for planning, damage assessment, and insurance purposes (Morris et al., 2018). Maps of groundwater flooding susceptibility are regularly constructed on the basis of lumped parameter models (Upton and Jackson, 2011) or by analyzing geological and hydrogeological data (British Geological Survey, 2006; Mancini et al., 2020; Yu et al., 2019). Recent studies employ this type of data with machine learning techniques to create spatially distributed maps of susceptibility to groundwater flood events (Allocca et al., 2021). Still, the lack of probabilistic and inverse methods in groundwater flooding models has been transferred to the groundwater flooding imagery, which often does not include formal likelihood quantification. This partially occurs due to the computational challenge from distributed models and high-dimensional inverse problems (Boyce and Yeh, 2014; Merz and Thielen, 2009; Xu et al., 2017; Zhou et al., 2014), and to limitations derived from data quality and availability (McKenzie et al., 2010; Merz and Thielen, 2009; Morris et al., 2018). Alternatives to overcome these problems are the reduction of the dimension of the input parameter space (Bittner et al., 2021; Boyce and Yeh, 2014; Erdal and Cırpka, 2019; Mara et al., 2017; Stanko et al., 2016; Teixeira Parente et al., 2019), and the optimization of the problem design by changing the spatial and temporal scaling (i.e., simulation period and model extension) and resolution (i.e., time and space discretization) (Guillaume et al., 2016; Scheibe et al., 2015). Assembling these alternatives can be useful to assess groundwater flooding in a probabilistic manner. As explained by Teng et al. (2017) and Di Baldassarre et al. (2010), probabilistic flood risks maps are preferable to single deterministic maps of inundation because they acknowledge the inherent uncertainty. In comparison to pluvial and fluvial flooding, there is less detail for hazard and risk management in groundwater flood mapping (Morris et al., 2018). As the demand for probabilistic maps in pluvial and fluvial flood risk analysis increases (Bhola et al., 2020; Teng et al., 2017), the need for groundwater flooding analysis is not expected to behave differently.

This study introduces a framework to recognize the susceptibility of residential property areas to groundwater flooding acknowledging the uncertainty in the model input parameters. We aim to create mapping products with the estimation of the probability of groundwater flood inundations by targeting one groundwater flood event as the basis of the problem design. We propose to decompose cell-by-cell the uncertainty and the sensitivity of a distributed groundwater numerical model to understand the spatial distribution and temporal variability of groundwater flood events. The framework involves three components: 1) elementary effects method to identify influential input factors and reduce the problem dimensionality, 2) Bayesian inference using the DiffereNTial Evolution Adaptive Metropolis (DREAM) algorithm to estimate statis-

tics of the relevant uncertain parameters and to quantify the output uncertainty incorporating evidence data, and 3) exploration of the predictive posterior distributions to assess the susceptibility to groundwater flooding according to exceedance probability levels. To demonstrate the relevance and feasibility of the application, the framework is applied to evaluate a groundwater flood event that occurred in the valley of the Alz River in Germany in 2013. The event is simulated using a numerical model built on MODFLOW-2005 (Harbaugh, 2005). The posterior probability distributions of the groundwater heads are then explored against groundwater depth thresholds and registered cellar depths.

## 2. Methods

### 2.1. Groundwater flow equation

Groundwater flow in a heterogeneous unconfined aquifer can be described by coupling the continuity equation with Darcy's law (Fetter, 1999), such that.

$$\frac{\partial}{\partial x} \left( k_{xx} \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left( k_{yy} \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left( k_{zz} \frac{\partial h}{\partial z} \right) + W = s \frac{\partial h}{\partial t}, \quad (1)$$

where  $h$  is the piezometric groundwater head [L],  $k_{xx}$ ,  $k_{yy}$ , and  $k_{zz}$  denote the hydraulic conductivity values [ $L T^{-1}$ ] along the  $x$ ,  $y$ , and  $z$  coordinate axes,  $s$  is the specific storage [ $L^{-1}$ ], and  $W$  represents the sources and sinks [ $T^{-1}$ ]. MODFLOW-2005 (Harbaugh, 2005) solves the groundwater flow equation in a finite difference form. The rate of change in storage within each cell is equal to the sum of flows into and out of the cell so that.

$$\sum Q_i = s \frac{\Delta h}{\Delta t} \Delta V, \quad (2)$$

where  $Q_i$  is the  $i$  component of the flow rate into the cell [ $L^3 T^{-1}$ ],  $\Delta V$  is the cell volume [ $L^3$ ], and  $\Delta h$  [L] is the change in the groundwater head over a time interval  $\Delta t$  [T]. External sources or stressors, such as the infiltration into the saturated zone and the river-aquifer fluxes, can be represented as  $Q_i$  elements. Hence, the forward problem is driven to predict the groundwater head distributions at successive times as a function of an uncertain set of model inputs  $\theta$  (i.e., uncertain parameters, source terms, initial or boundary conditions).

### 2.2. Sensitivity analysis

Sensitivity analysis is used to quantify how the uncertainty in the output of a model can be apportioned to the different uncertainties in the model inputs (Saltelli, 2002; Smith, 2013). We use the elementary effects method, also known as Morris method, to identify the sensitive parameters through basic statistical moments (Campolongo et al., 2007; Morris, 1991). Due to its sampling strategy, this technique is advantageous for quantifying the sensitivity in computationally expensive models with many uncertain inputs (Campolongo et al., 2007; Jaxa-Rozen and Kwakkel, 2018; Smith, 2013). In our research, the purpose of its application is to identify which input factors are non-influential and therefore negligible, as well as to recognize the input factors that are non-linear or involved in interactions. Those input factors that are non-influential are fixed to reduce the problem dimensionality, which is an essential step to define an affordable experimental design and proceed with the subsequent Bayesian analysis.

The elementary effect method considers that each model input  $q = [q_1, \dots, q_k]$  varies across  $p$  selected levels, forming a  $k$ -dimensional  $p$ -level finite grid of experimentation  $\Gamma_p$ . Then,  $p$  represents the number of partitions of the model parameter space at which the model can be evaluated. Assuming that the input terms are scaled to the interval  $[0, 1]$ , the elementary effect associated with the input  $q_i$  is defined as.

$$d_i(q) = \frac{f(q_1, \dots, q_{i-1}, q_i + \Delta, q_{i+1}, \dots, q_k) - f(q)}{\Delta} \quad (3)$$

$$= \frac{f(q + \Delta e_i) - f(q)}{\Delta},$$

with a jump size  $\Delta \in \{1/(p-1), \dots, 1-1/(p-1)\}$  and transformed point  $q + \Delta e_i$ , where  $q = (q_1, \dots, q_k)$  is any selected value in  $\Gamma_p$ , and  $e_i$  is a vector of zeros with one in the  $i^{\text{th}}$  component. Then,  $d_i(q)$  quantifies the local sensitivity behavior at the point  $q$ . The global sensitivity measure requires to approximate the statistical moments from the associated finite-dimensional distribution  $G_i$  by randomly sampling  $q$  from the grid points in  $\Gamma_p$ . Morris (1991) proposed the mean  $\mu_i$  and the standard deviation  $\sigma_i$  of the distribution  $G_i$ . By sampling  $m$  points, these metrics associated with the  $i^{\text{th}}$  parameter and  $j^{\text{th}}$  samples can be expressed as.

$$\mu_i = \frac{1}{m} \sum_{j=1}^m d_i^j(q), \quad (4)$$

$$\sigma_i = \sqrt{\frac{1}{m-1} \sum_{j=1}^m (d_i^j(q) - \mu_i)^2}. \quad (5)$$

Campolongo et al. (2007) introduced an additional sensitivity metric  $\mu_i^*$  to avoid the problems associated to non-monotonic models that may contain effects with opposite signs. This refined version estimates the mean of the absolute values of the elementary effects and can be written as.

$$\mu_i^* = \frac{1}{m} \sum_{j=1}^m |d_i^j(q)|. \quad (6)$$

The elementary effects method proposes to construct  $\mu_i^*$  and  $\sigma_i$  with  $m$  trajectories of  $(k+1)$  points in the input space, which provide  $k$  elementary effects. The total number of realizations is defined by  $(k+1) \times m$ . The sampling strategy employs the orientation matrix  $\mathbf{B}^*$  to obtain random samples from  $G_i$ , so that.

$$\mathbf{B}^* = \left( \mathbf{J}_{k+1,1} q^* + \frac{\Delta}{2} [(2\mathbf{B} - \mathbf{J}_{k+1,k}) \mathbf{D}^* + \mathbf{J}_{k+1,k}] \right) \mathbf{P}^*, \quad (7)$$

where  $q^*$  is a random initial point,  $\mathbf{J}_{k+1,k}$  is a  $(k+1) \times k$  matrix of ones,  $\mathbf{B}$  is a  $(k+1) \times k$  strictly lower triangular matrix of ones,  $\mathbf{D}^*$  is a  $k \times k$  diagonal matrix formed by randomly chosen elements from the set  $\{-1, 1\}$ , and  $\mathbf{P}^*$  is a  $k \times k$  random permutation matrix.

## 2.3. Bayesian inference and sampling algorithm

### 2.3.1. Bayesian inference

Bayesian inference is derived from the Bayes' theorem to explicitly introduce in the analysis assumptions about evidence data and prior knowledge of the model inputs (Box and Tiao, 2011; Smith, 2013). Let  $\mathcal{Y}$  represent a given set of observations or evidence data and  $\theta$  denote a set of parameters. Following the Bayesian formulation, the posterior probability of the parameter set  $P(\theta|\mathcal{Y})$  can be obtained with the prior probability of the parameter set  $P(\theta)$ , the observed data  $P(\mathcal{Y})$ , and the likelihood function  $P(\mathcal{Y}|\theta) \equiv L(\theta|\mathcal{Y})$ , such that.

$$P(\theta|\mathcal{Y}) = \frac{P(\mathcal{Y}|\theta) \times P(\theta)}{P(\mathcal{Y})}. \quad (8)$$

Given that all statistical inferences of  $P(\theta|\mathcal{Y})$  can be made from the unnormalized density,  $P(\mathcal{Y})$  can be omitted (Vrugt, 2016), and the Bayes equation is simplified as.

$$P(\theta|\mathcal{Y}) \propto L(\theta|\mathcal{Y}) \times P(\theta). \quad (9)$$

$L(\theta|\mathcal{Y})$  summarizes the distance between the observation and the simulated quantity of interest (groundwater heads for the sake of our research). Our formulation of the Bayesian inference problem assumes that the error residuals are homoscedastic and normally distributed  $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ , and it employs the log-likelihood  $\mathcal{L}(\theta|\mathcal{Y})$  for numerical stability (Vrugt, 2016). Then, considering  $n$ -vector of error residuals at times  $t_s = \{1, \dots, n\}$ , the log-likelihood function applied in this study can be written as.

$$\mathcal{L}(\theta|\mathcal{Y}) = -\frac{n}{2} \times \log \left[ \sum_{t_s=1}^n \varepsilon_t(\theta)^2 \right]. \quad (10)$$

For further debate about mathematical definitions of the likelihood function and details in their derivation process, we refer the readers to the corresponding literature (Kaipio and Somersalo, 2005; Schoups and Vrugt, 2010; Vrugt, 2016).

### 2.3.2. Sampling algorithm

To estimate the probability density function of the parameters, Markov Chain Monte Carlo (MCMC) sampling is implemented using the differential evolution adaptive metropolis (DREAM) scheme (Vrugt et al., 2009, 2008; Vrugt and Ter Braak, 2011). In general, MCMC techniques consist of the iterative sampling of a probabilistic space to find areas of a high likelihood of the posterior distribution of the parameter space. The DREAM algorithm is an efficient MCMC-based method. It runs multiple chains of simulations simultaneously and adjusts the scale and orientation of the proposal distribution during the run. Candidate points are created by sampling the prior parameter distribution for each chain, and the Metropolis-Hastings ratio is used to decide if a candidate point is accepted or not. The convergence of each chain is evaluated using the Gelman-Rubin diagnostic (Gelman and Rubin, 1992). Extensive theoretical descriptions of the DREAM algorithm implementation can be found in the literature (Houska et al., 2015; Vrugt, 2016; Vrugt et al., 2009, 2008; Vrugt and Ter Braak, 2011).

## 2.4. Exceedance probability and susceptibility maps

The final step of the framework involves extracting information from the posterior distributions of the groundwater heads at each element of the model and at different time steps to analyze the spatial and temporal conditions of the event. We characterize the associated groundwater flooding risk by computing the probability of the groundwater table of exceeding a certain threshold value (i.e., exceedance probability). For convenience, this can be explained from the cumulative distribution function of the predicted groundwater heads  $F(h)$  for  $F: \mathbb{R} \rightarrow [0, 1]$ , that can be described by.

$$F(h) := P(H \leq h). \quad (11)$$

The exceedance probability  $\bar{P}$  can then be estimated with the complementary cumulative distribution function of Eq. (11), expressed as.

$$\bar{P} = P(H > h) = 1 - F(h). \quad (12)$$

Consequently, we create the maps of susceptibility to groundwater flooding considering the probability  $\bar{P}_{ij}^{(t)}$  of exceeding a threshold  $K_{ij}$ , computed at the time step  $t = [t_1, \dots, t_n]$  and the discrete point  $i = 1, \dots, n_c$  and  $j = 1, \dots, n_r$ , where  $n_c$  and  $n_r$  are, respectively, the number of columns and rows used for the spatial discretization. Thus,

$$\bar{P}_{ij}^{(t)} = 1 - F_{ij}^{(t)}(K_{ij}) \quad (13)$$

We propose two types of thresholds in this work. The first one is the groundwater depth threshold  $K_{ij}$ , which represents a specified distance  $d$  between the surface elevation and the groundwater table at the cell

$i, j$ . Hence, it can be defined according to a two-dimensional array of a digital elevation model  $T_{ij}$ , that is,

$$\kappa_{ij} = T_{ij} - d. \quad (14)$$

The second threshold is defined according to the cellar depths  $\hat{\kappa}_{ij}$  at a specific house, property, or building. Hence,  $\hat{\kappa}_{ij}$  is the reported depth of the underground infrastructure located at the cell  $i, j$ . Therefore,  $\kappa_{ij}$  can be equal to  $\kappa_{ij}$  or  $\hat{\kappa}_{ij}$  depending on the assessment.

## 2.5. Practical application

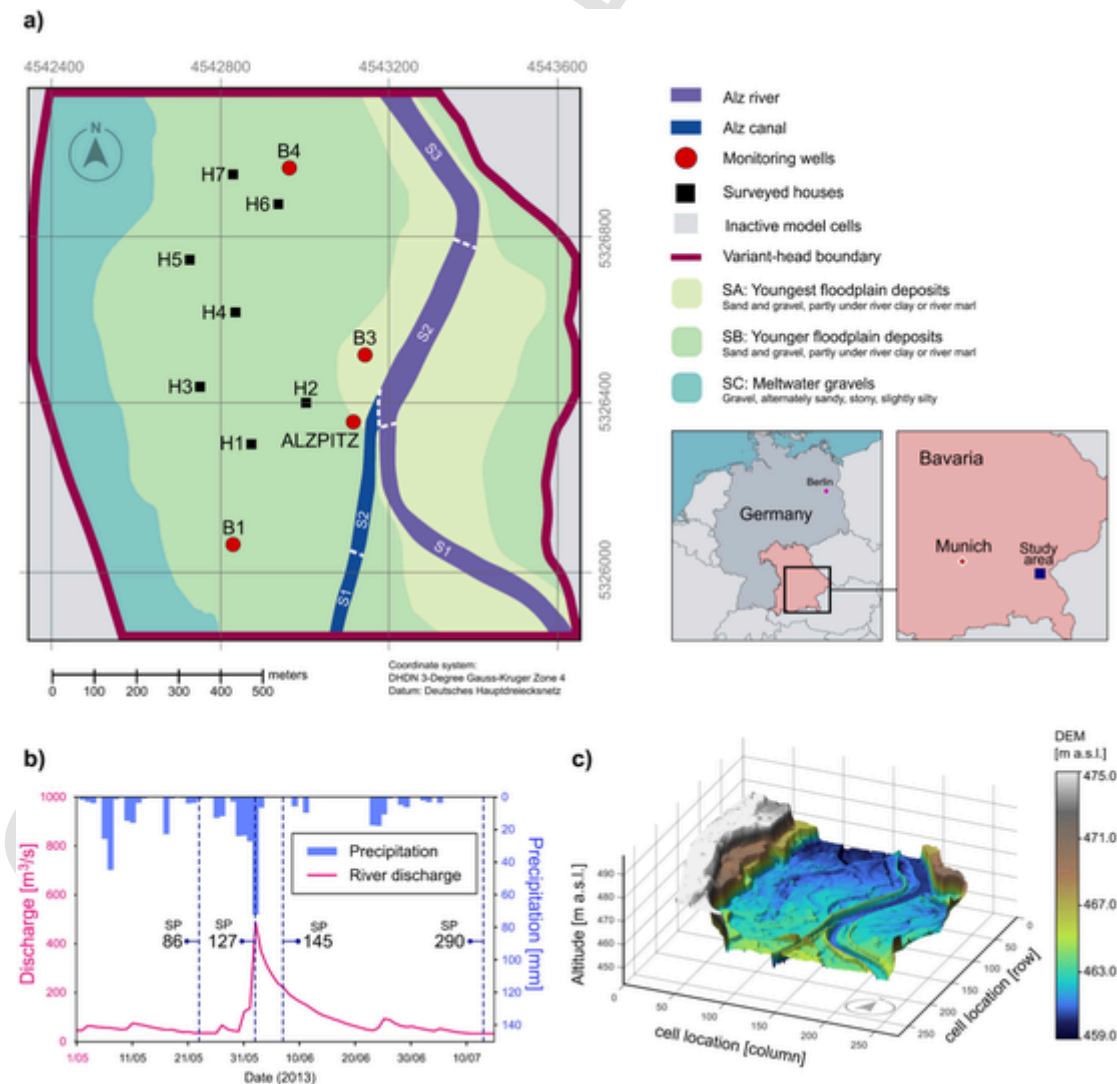
### 2.5.1. Case study

The study area is located in southeastern Bavaria, Germany. During heavy rainfall events at the beginning of June 2013, the region was affected by flood events that caused several damages in the residential properties (Disse et al., 2015; Keilholz et al., 2015). The study region is characterized by an unconsolidated shallow sedimentary aquifer formed by gravels and sands (Doppler et al., 2011; Keilholz et al., 2015; Merchán-Rivera et al., 2018). The groundwater flow is very dynamic due to the high permeability of the porous material of the aquifer, the relatively rapid groundwater recharge, and the hydraulic connection

with the Alz River and the Alz Canal. The flood event in 2013 in the Alz River valley and the hydrogeological features of the region were the subjects of previous studies. Doppler et al. (2011) and Bayerisches Landesamt für Umwelt (2017) describe the geological properties and the morphological features of the valley. Further studies also estimate the uncertainty in the discharge and stage of the Alz River (Willems, 2011; Willems and Stricker, 2012) and quantify the propagation of uncertainty due to the representation of the river boundary conditions (Merchán-Rivera et al., 2021). Fig. 1 includes details of the study area and the flood event.

### 2.5.2. Model description

A two-dimensional model was built using MODFLOW-2005 (Harbaugh, 2005) to describe the event. The model simulates 300 stress periods at intervals of six hours to cover a total of 75 days (from May 02, 2013 to July 15, 2013). The spatial discretization is a finite-difference grid formed by one layer, 260 rows, and 260 columns. The initial and boundary conditions were defined based on the preexisting regional model built by Keilholz (2015). Three types of boundary conditions are used in the model: 1) specified head boundaries, using time-variant specified head package (Harbaugh et al., 2000; Leake and Prudic, 1991) to simulate the variation in the water table at the borders



**Fig. 1.** Study case implementation: a) study area location and model setting with groundwater monitoring wells (circles) and the selected surveyed houses (squares); b) event description and representative stress periods (SP) for the different event phases; and, c) digital elevation model (DEM) used to represent the surface elevation.



of the domain, 2) recharge into the saturated zone, using the recharge package (Harbaugh, 2005; Harbaugh et al., 2000) to simulate the distributed infiltration flux, and 3) river interactions, using the river package (Harbaugh, 2005; Harbaugh et al., 2000) to simulate the interaction between the streams and the aquifer. The spatial distribution of the specified head and river boundaries are illustrated in Fig. 1a. Given that the initial conditions of the groundwater heads and the specified-head boundaries were extracted from the regional model (Keilholz et al., 2015), which include a larger simulation period, there was no need to include a warm up period. Three hydraulic conductivity values ( $k_A$ ,  $k_B$ , and  $k_C$ ) are defined based on the geological studies (Bayerisches Landesamt für Umwelt, 2017; Doppler et al., 2011), and, accordingly, three specific storage values ( $s_A$ ,  $s_B$ , and  $s_C$ ) and three specific yield values ( $\gamma_A$ ,  $\gamma_B$ , and  $\gamma_C$ ). These parameters were assumed to be homogeneous within each soil section (SA, SB and SC). Recharge into the saturated zone values ( $Q_{ij}^{(t)}$ ) are defined according to the deterministic estimations from the hydrological model made by Keilholz et al. (2015). Finally, the streams are parameterized considering two streambed conductance values for the canal ( $c_1$  and  $c_2$ ), three streambed conductance values for the river ( $r_1$ ,  $r_2$ , and  $r_3$ ), which are outlined according to observed differences in the morphology of the streams, and stream stage values ( $\xi_{ij}^{(t)}$ ) based on the previous estimations made by Merchán-Rivera et al. (2021). Groundwater field measurements were registered every six hours during the simulation period in four groundwater monitoring wells: Alzpitz, B1, B3, and B4 (Keilholz et al., 2015). These observations were used as the evidence for the Bayesian inference. Although we computed the hydraulic head values for the entire simulation period, we chose four simulation time steps to be representative of different phases of the event to present the results: before the event at stress period 86 (May 23, 2013), peak-flow at stress period 127 (June 02, 2013), recession phase at stress period 145 (June 07, 2013), and after the flood event at stress period 290 (July 13, 2013). We define the recession phase as the stage that immediately follows the peak-flow, whereas after the event refers to the phase where the declining period has ended. These separate periods were used along with the results and discussion to understand the temporal variation of the outcomes. Fig. 1 includes details of the study, the model settings and the digital elevation model.

### 2.5.3. Scheme implementation

The hydraulic properties are summarized in Table 1, where  $\mathcal{U}(a, b)$  denotes the prior belief distributed uniformly between  $a$  and  $b$ . Except for  $Q_{ij}^{(t)}$  and  $\xi_{ij}^{(t)}$ , the uncertainty analysis considers the parameters as

**Table 1**  
Distribution properties of the stochastic input variables.

| Parameter                | Zone | Notation   | Distribution                    | Units             |
|--------------------------|------|------------|---------------------------------|-------------------|
| Hydraulic conductivity   | SA   | $k_A$      | $\mathcal{U}(1e^{-4}, 1e^{-1})$ | m/s               |
|                          | SB   | $k_B$      | $\mathcal{U}(1e^{-4}, 1e^{-1})$ | m/s               |
|                          | SC   | $k_C$      | $\mathcal{U}(1e^{-6}, 1e^{-3})$ | m/s               |
| Specific storage         | SA   | $s_A$      | $\mathcal{U}(1e^{-7}, 1e^{-3})$ | 1/s               |
|                          | SB   | $s_B$      | $\mathcal{U}(1e^{-7}, 1e^{-3})$ | 1/s               |
|                          | SC   | $s_C$      | $\mathcal{U}(1e^{-7}, 1e^{-3})$ | 1/s               |
| Specific yield           | SA   | $\gamma_A$ | $\mathcal{U}(0.10, 0.40)$       | —                 |
|                          | SB   | $\gamma_B$ | $\mathcal{U}(0.10, 0.40)$       | —                 |
|                          | SC   | $\gamma_C$ | $\mathcal{U}(0.10, 0.40)$       | —                 |
| Recharge multiplier      | —    | $R$        | $\mathcal{U}(0.00, 2.00)$       | —                 |
| Conductance of the canal | S1   | $c_1$      | $\mathcal{U}(1e^{-5}, 9e^{-1})$ | m <sup>2</sup> /s |
|                          | S2   | $c_2$      | $\mathcal{U}(1e^{-5}, 9e^{-1})$ | m <sup>2</sup> /s |
| Conductance of the river | S1   | $r_1$      | $\mathcal{U}(1e^{-7}, 9e^{-4})$ | m <sup>2</sup> /s |
|                          | S2   | $r_2$      | $\mathcal{U}(1e^{-7}, 9e^{-4})$ | m <sup>2</sup> /s |
|                          | S3   | $r_3$      | $\mathcal{U}(1e^{-7}, 9e^{-4})$ | m <sup>2</sup> /s |
| Stream stage variation   | —    | $\psi$     | $\mathcal{U}(-0.145, 0.145)$    | m                 |

random variables that are simply sampled from a uniform distribution and used in the realizations of the scheme. The choice to set the prior beliefs to be uniform (i.e., non-informative) is a conservative assumption at costs of computational efficiency. In practice, this means that we need a larger sample size to obtain better estimates of the posterior distribution during Bayesian inversion. More informative prior distributions would require a smaller sample size to reach a similar precision. Note that the sampling strategy in the elementary effect method should be modified if non-uniform distributions are considered (Feng et al., 2019).

The recharge influx into the saturated zone  $Q_{ij}^{(t)}$  is spatially distributed over the domain and varies at every stress period  $t$  of the simulation. Keilholz et al. (2015) estimated deterministic values for recharge into the saturated zone  $\tilde{Q}_{ij}^{(t)}$  in a regional hydrologic model using MIKE SHE (DHI, 2013). In our research, the uncertainty in these recharge values is introduced as a single multiplier  $R$  for all the grid cells affected by recharge at all time steps, similarly to the approach employed by Mustafa et al. (2018). Hence, the random variable  $R \sim \mathcal{U}(a, b)$  is used in combination with  $\tilde{Q}_{ij}^{(t)}$  to obtain the corrected spatial distributed recharge  $Q_{ij}^{(t)}$ , such that.

$$Q_{ij}^{(t)} = \tilde{Q}_{ij}^{(t)} \times R. \quad (15)$$

Similarly, Keilholz et al. (2015) provided scatter stage information for the river and the canal. These scatter data were post-processed by Merchán-Rivera et al. (2021) to obtain the stream head  $\tilde{\xi}_{ij}^{(t)}$  at every cell that represents the river and the canal. Hence, the streams are represented as time-variant and spatially distributed inputs. The uncertainty in the stage is included by the error  $\psi \sim \mathcal{U}(a, b)$ , where the bounds  $a$  and  $b$  are approximated according to the works by Willems (2011) and Willems and Stricker (2012). This means that  $\psi$  increases or reduces the stream stage  $\tilde{\xi}_{ij}^{(t)}$ , for all of the stress periods  $t$  at each cell  $i, j$  that represent the streams to obtain.

$$\xi_{ij}^{(t)} = \tilde{\xi}_{ij}^{(t)} + \psi. \quad (16)$$

The implementation of the elementary effects considers all parameters mentioned in Table 1 as input factors. Given that the values of  $p$  and  $m$  are primarily an experimental choice (Saltelli et al., 2004), we performed an extensive exploration of different factor levels and sampling size by constructing various experimental combinations using  $p = 4, 6, 8$  and  $10$ , and  $m = 10, 20$  and  $30$ . The elementary effects approach reduces the dimensionality of our problem and provide a list of influential factors that are later used in the Bayesian inversion with the DREAM algorithm. The posterior distributions are calculated from the last 50% of the samples generated with the DREAM algorithm from 10,000 realization. We explore the posterior predictive distributions of the groundwater heads at every element of the domain and every stress period. To compute the threshold  $\kappa_{ij}$ , we define  $T_{ij}$  with the digital elevation model (DEM) of the region (Bayerisches Landesamt für Umwelt, 2017) and the groundwater depth  $d$  to be equal to 0.5 and 1.5. Note that  $d$  values are arbitrarily selected, and they can simply be modified once the posterior predictive distributions are quantified. The DEM resolution is  $5 \times 5$  m cell size. Finally, we define the threshold  $\hat{\kappa}_{ij}$  using the cellar depth records from the household survey run by Disse et al. (2015) in April 2014:  $\hat{\kappa}_{H1} = 459.0$  m a.s.l. (above sea level),  $\hat{\kappa}_{H2} = 458.6$  m a.s.l.,  $\hat{\kappa}_{H3} = 459.4$  m a.s.l.,  $\hat{\kappa}_{H4} = 461.1$  m a.s.l.,  $\hat{\kappa}_{H5} = 460.1$  m a.s.l.,  $\hat{\kappa}_{H6} = 461.0$  m a.s.l.,  $\hat{\kappa}_{H7} = 460.3$  m a.s.l.

We employ the SALib Python library (Herman and Usher, 2017) to compute the elementary effect indices with the efficient sampling strategy introduced by Campolongo et al. (2007). The DREAM algorithm was implemented using the SPOTPY framework (Houska et al., 2015).

The probability maps in the results were created by arranging the output data in matrix arrays using the Numpy library (Harris et al., 2020) and plotting them with the Matplotlib library (Hunter, 2007) in Python 3. The algorithm for implementing the exposed approach was written in Python 3, and the files are available in the research dataset (Merchán-Rivera, 2021).

### 3. Results and discussion

#### 3.1. Sensitivity analysis

Based on the values of  $\mu^*$ , we obtain a ranking of model sensitivity that can be easily interpreted (see Fig. 2). A high absolute measure of

central tendency  $\mu^*$  denotes an input with a significant overall influence on the output. Hence, the most sensitive parameters are  $k_A$ ,  $k_B$ ,  $\gamma_A$ ,  $R$ ,  $c_2$ , and  $r_2$ . Also, there are spatial and temporal variabilities in the results of the elementary effects. The positions in the ranking change depending on the location and the phase of the event. We observe higher values of the elementary effects metrics at the peak-flow. Similarly, we can see more significant differences between the minimum and maximum values of  $\mu^*$  at this phase. The spatial variation in the sensitivity of the recharge multiplier  $R$  occurs due to the heterogeneous distribution of the recharge into the saturated zone  $\tilde{Q}_{ij}^{(n)}$ . The parameters that describe the streams ( $c_1$ ,  $c_2$ ,  $r_1$ ,  $r_2$ ,  $r_3$ , and  $\psi$ ) gain relevance close to the streams at the monitoring wells Alzpitz and B3. The sensitivity of the canal conductance  $c_2$  is also large at B1 and H1.

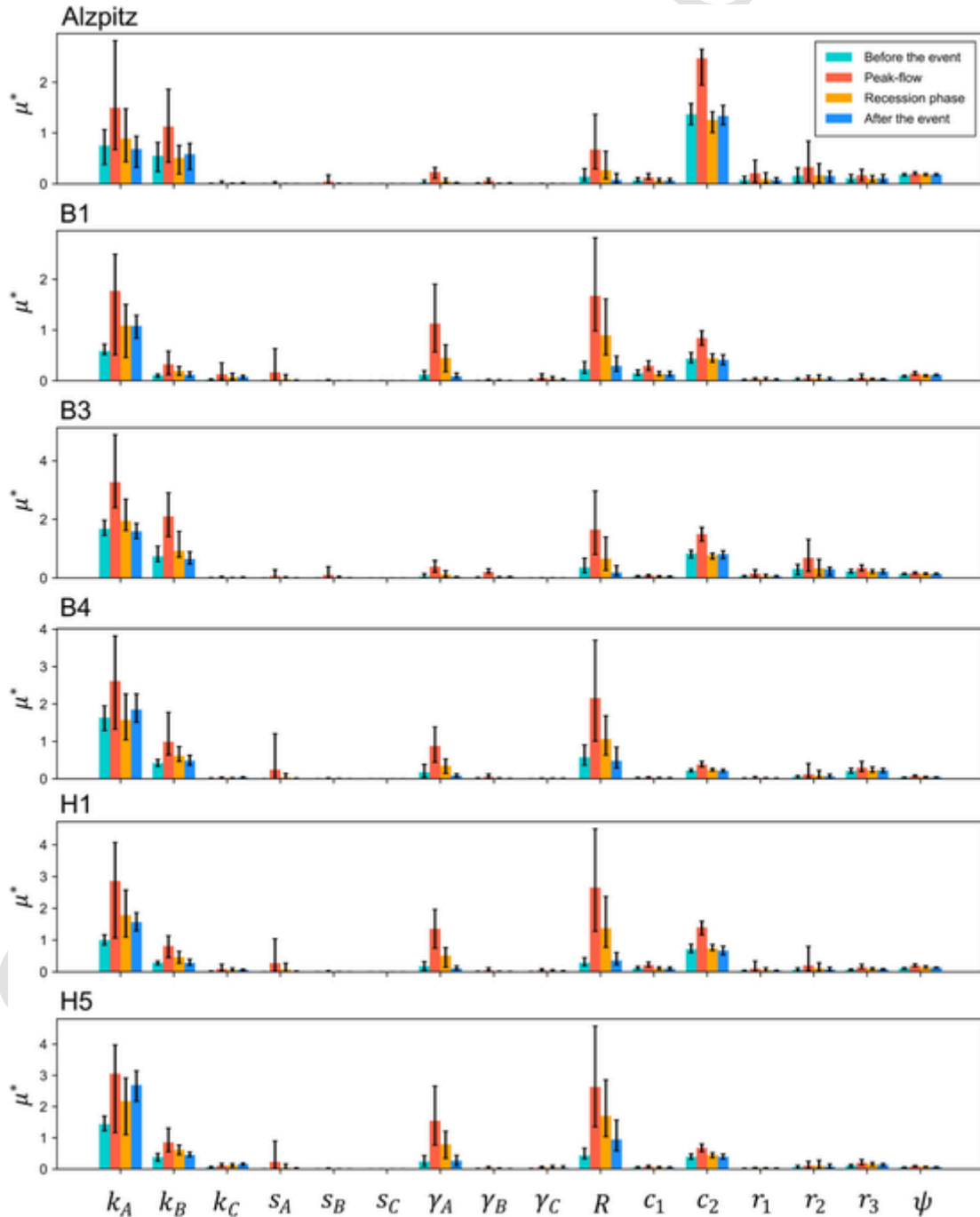


Fig. 2. Elementary effects results per parameter at different location. The colored bars represent the mean value of  $\mu^*$  at the different phases of the event. The upper end and lower end of the black whiskers show the maximum and minimum value of  $\mu^*$ , respectively.

The second sensitivity measure that we obtain is the spread  $\sigma$ . A high value of  $\sigma$  indicates input factors with a non-linear effect in the model output, the existence of parameter interactions, or both. The ratio  $\sigma/\mu^*$  is an indicator of model linearity (if  $\sigma/\mu^* = 0$ ), or non-linearity (if  $\sigma/\mu^* > 0$ ). Also, this indicator can occasionally give evidence of monotonic and non-monotonic model responses, when  $0 < \sigma/\mu^* < 0.5$  and  $\sigma/\mu^* > 0.5$ , respectively (García Sánchez et al., 2014). In Fig. 3, we display all the outcomes of  $\mu^*$  and  $\sigma$  from every combination of  $p$  and  $m$  and classify them based on the event phase in a scatter plot. Overall, the model behavior is subject of non-linearities, non-monotonicities, or in-

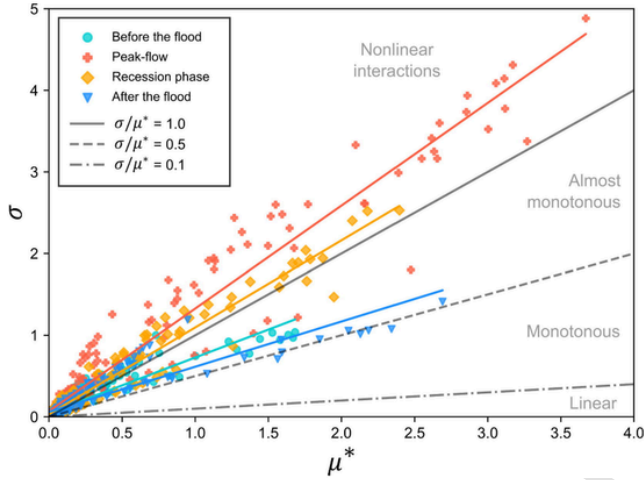


Fig. 3. Covariance of  $\mu^*$  and  $\sigma$  at different phases of the flood event. Colored lines show a simple linear regression to illustrate the responses at the different phases. Gray lines delimit the referential areas that describe nonlinear, almost monotonous, monotonous and linear interactions, according to the ratio  $\sigma/\mu^*$ .

teractions between model parameters. The results also show temporal variation in this behavior. While the functional relationships between the input factors and the outputs are highly non-linear, interactive, or non-monotonic during the peak and the recession phase, the relationships tend to reduce these qualities before and after the event. Non-monotonic responses can occur with respect to the input factors associated to the hydraulic conductivity, particularly  $k_A$  and  $k_B$ . The effect of the parameter interactions can increase due to the increased influx from the high staged-river seen in the peak-flow and the recession phase, which would also influence the non-monotonicity from  $k_A$  and  $k_B$ . Notice that the non-linearity of the model responses is not affected directly by variations in the influx from the streams and the recharge but by the effect that these have on the hydraulic gradient magnitude. Five parameters can be considered non-influential factors:  $k_C$ ,  $s_A$ ,  $s_B$ ,  $s_C$ , and  $\gamma_C$ . As expected, the specific storage values are negligible because we study an unconfined aquifer and their contribution to the storativity is orders of magnitude lower than the specific yield. In the case of  $k_C$  and  $\gamma_C$ , the low values of  $\mu^*$  are determined by the spatial location of the soil type SC. Hence, these five parameters are fixed to reduce the problem dimensionality for the Bayesian inversion from 16 to 11 calibration parameters. Even if a further reduction of the parameter space could be assumed and be quite useful for the application of the Bayesian inversion, we want to retain the properties of the original model for the scope of the study at the cost of higher computational demand.

### 3.2. Bayesian inversion

In Fig. 4, we can observe the prior and posterior discrete distributions of the sensitive parameters. These results show that the algorithm updated the distributions of the sensitive parameters, reducing their uncertainty. We also observe that most of the parameters tend to unimodality, except  $\psi$ , whose posterior distribution is only slightly up-

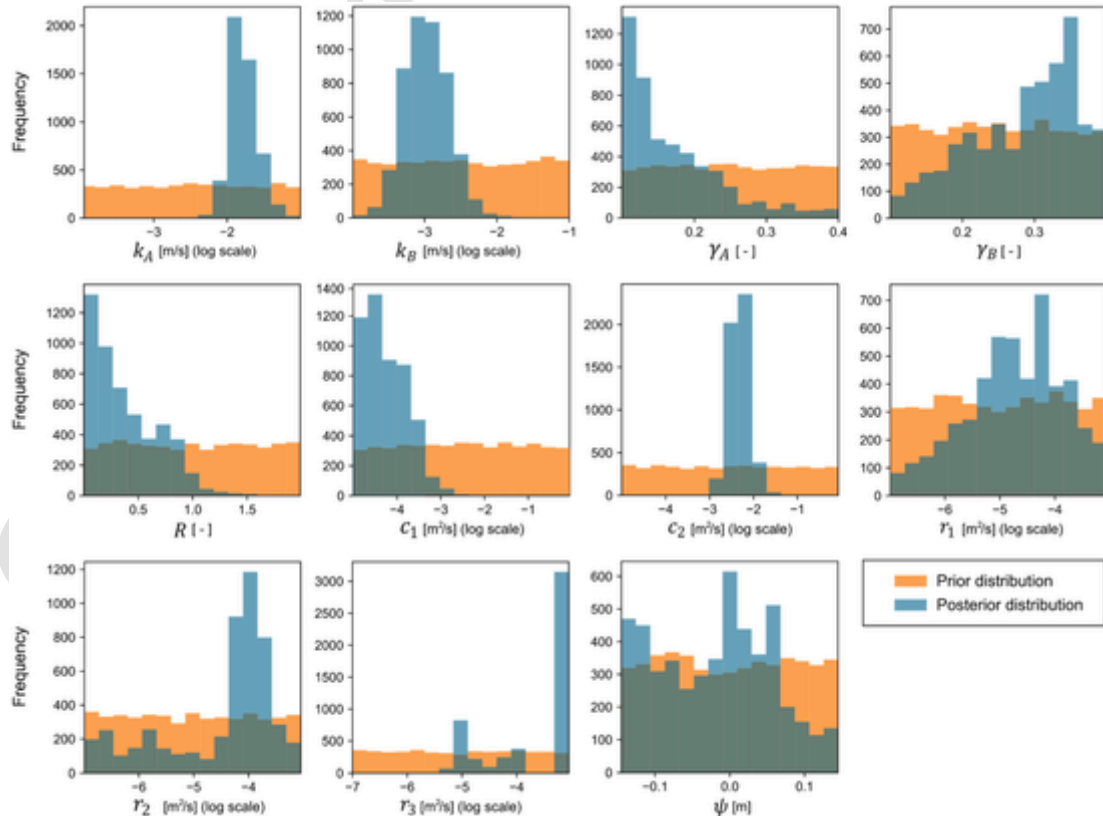


Fig. 4. Prior and posterior discrete representation of the distributions for the classified sensitive parameters. The orange histogram shows the prior parameter distribution sampled from the Halton uniform distribution, while the blue histogram shows the posterior parameter distribution.

dated in comparison to the prior distribution. This can be reasonably explained by the sensitivity analysis that shows low scores for this parameter. The statistical moments of the posterior distribution of the uncertain parameters can be found in Table 2. The expected values of the conductance from the river sections  $r_1$  and  $r_2$  range within the same order of magnitude. The latter suggests that the bed conductance of the river may be overparameterized, considering the location of the observation wells used for the inversion. Given these results and the outcomes of the sensitivity analysis, an additional reduction of the model dimensionality could be possible by grouping  $r_1$  and  $r_2$  in one parameter. We observe a high frequency of samples in the lower end of  $R$  that reduces the revised infiltration  $Q_{ij}^{(r)}$  used in the model evaluations. This implies that the deterministic values of the infiltration into the saturated zone extracted from the regional model  $\tilde{Q}_{ij}^{(r)}$  do not represent accurately the system, whether by overestimating the inflow rates or the recharge extension. This remarks the importance of quantifying the uncertainty in the analysis of risk of these events. Still, note that  $R$  and  $\psi$  are random variables that aim to catch the uncertainty in model parameters that are fundamentally transient: recharge into the saturated zone  $Q_{ij}^{(r)}$  and river stage  $\xi_{ij}^{(r)}$ . This means that a single random sample per evaluation may not exhaustively capture the temporal variability and the uncertainty of these parameters. Also, it is important to point out that the Bayesian approach relies in the information fed by the empirical evidence. Therefore, the location of the observation wells plays a fundamental role. Some parameters may not be adequately represented due to the location of the monitoring wells, such as  $R$ ,  $c_1$ ,  $r_1$  and  $r_3$ .

The reduction of the model parametric uncertainty is evident after taking into consideration the observed data with the Bayesian approach (see Table 3 and Fig. 5). In Fig. 5, we can see the probability distribution of the groundwater heads considering the prior and posterior distribution of the sensitive parameters. A lower standard deviation  $\sigma_o$  in the hydraulic head distribution indicates that the values tend to be

**Table 2**

Statistical moments of the posterior parameter distributions (mean  $\mu_\theta$ , the standard deviation  $\sigma_\theta$ , the skewness  $\gamma_\theta$ , and kurtosis  $\beta_\theta$ ), and best set of parameters.

| Parameter  | Parameter units                 | $\mu_\theta$ | $\sigma_\theta$ | $\gamma_\theta$ | $\beta_\theta$ | Best set |
|------------|---------------------------------|--------------|-----------------|-----------------|----------------|----------|
| $k_A$      | [m/s] (log scale)               | -1.7742      | 0.3093          | -1.1121         | 5.0517         | -1.9021  |
| $k_B$      | [m/s] (log scale)               | -2.8872      | 0.3952          | 0.4756          | 1.0179         | -3.2105  |
| $\gamma_A$ | [ - ]                           | 0.2111       | 0.0858          | 0.4965          | -0.9822        | 0.1015   |
| $\gamma_B$ | [ - ]                           | 0.2794       | 0.0783          | -0.4364         | -0.8639        | 0.1866   |
| $R$        | [ - ]                           | 0.5662       | 0.4745          | 0.9019          | 0.0115         | 0.1304   |
| $c_1$      | [m <sup>2</sup> /s] (log scale) | -3.9211      | 0.9285          | 1.6027          | 2.7368         | -4.8203  |
| $c_2$      | [m <sup>2</sup> /s] (log scale) | -2.3905      | 0.7497          | -1.2957         | 3.8439         | -2.3908  |
| $r_1$      | [m <sup>2</sup> /s] (log scale) | -4.7159      | 1.0344          | -0.3018         | -0.8592        | -5.1017  |
| $r_2$      | [m <sup>2</sup> /s] (log scale) | -4.5951      | 1.0739          | -0.747          | -0.7445        | -4.2346  |
| $r_3$      | [m <sup>2</sup> /s] (log scale) | -4.005       | 1.0621          | -0.898          | -0.2509        | -3.0910  |
| $\psi$     | [m]                             | -0.0225      | 0.0791          | 0.1594          | -1.0402        | -0.0924  |

**Table 3**

Statistical moments of the prior and posterior predictive uncertainty (mean  $\mu_o$  and standard deviation  $\sigma_o$ ) at the different monitoring wells during different phases of the event.

| Flood phase      | Dist.     | Alzpitz  |            | B1       |            | B3       |            | B4       |            |
|------------------|-----------|----------|------------|----------|------------|----------|------------|----------|------------|
|                  |           | $\mu_o$  | $\sigma_o$ | $\mu_o$  | $\sigma_o$ | $\mu_o$  | $\sigma_o$ | $\mu_o$  | $\sigma_o$ |
| Before the flood | Prior     | 461.7874 | 1.1424     | 461.702  | 1.0063     | 461.1344 | 1.1659     | 458.4919 | 0.9965     |
|                  | Posterior | 461.8520 | 0.2225     | 461.5965 | 0.0923     | 460.7141 | 0.1851     | 458.1167 | 0.1211     |
| Peak-flow        | Prior     | 463.4296 | 1.5545     | 462.9069 | 1.3847     | 462.6298 | 1.8316     | 459.4000 | 1.4593     |
|                  | Posterior | 463.4992 | 0.3978     | 462.7843 | 0.1890     | 461.8404 | 0.3655     | 458.9123 | 0.1704     |
| Recession phase  | Prior     | 462.7484 | 1.1706     | 462.6818 | 1.1090     | 462.1369 | 1.2520     | 459.3888 | 1.0565     |
|                  | Posterior | 462.7654 | 0.2065     | 462.5491 | 0.0951     | 461.6666 | 0.1879     | 459.0749 | 0.1191     |
| After the flood  | Prior     | 461.6083 | 1.1259     | 461.7917 | 1.0442     | 460.9604 | 1.1409     | 458.4946 | 1.0216     |
|                  | Posterior | 461.6909 | 0.2146     | 461.5675 | 0.0938     | 460.5780 | 0.1749     | 458.0773 | 0.1225     |

close to the central tendency metric  $\mu_o$  (see Table 3). Fig. 5 shows the approximation of the probability density function of the groundwater heads from the prior and the posterior predictive simulations at different locations and different phases of the event. The kernel density plot was calculated using Scott's Rule (Scott, 2010) over the finite output data of the groundwater heads located at the different monitoring wells. This plot is helpful to identify the position and relative amplitude of the density distributions and to recognize if the distribution is multimodal. The groundwater heads at Alzpitz, located immediately next to the river, display a bimodal distribution with the parameterization of the model using the prior distribution of the parameters. This distribution is later updated with the posterior to a unimodal distribution. The spread of the groundwater heads is evidence of significant uncertainty in the prior beliefs, which is driven by the interconnection with the river, especially during the peak-flow discharges. The interquartile range between the first and third quartile (Q3 – Q1 range) shows the variability around the median of 50% of the predictive values of the groundwater heads using the prior distribution of the parameters. We see that the Q3 – Q1 range is larger than one meter at Alzpitz and B3 for the model considering the prior distribution of the parameters. Lastly, we can observe the skewness of the prior predictive distributions, which depict positive asymmetry about their mean. Fig. 6 shows the final results of the predictive groundwater heads after applying the Bayesian inference. We use the mean and the interval  $[\mu_o + \sigma_o, \mu_o - \sigma_o]$  of the predictive posterior distribution as a measure of the uncertainty. We observe that these metrics properly capture the observations, particularly in the wells Alzpitz and B3. At B4, the predictive uncertainty does not include the peak observations. Also, we observe disagreements in the responses at B1 that may occur due to the influence of the southern time-variant specified heads boundary conditions.

### 3.3. Probability maps

The Bayesian approach allows us to study groundwater flood events considering the posterior distribution of the groundwater levels, which can be used for subsequent analysis, such as transport modeling, scenario modeling, or assembling more complex model structures. In our case, the posterior distributions at every cell of the domain can be used to create probability maps of susceptibility to groundwater flooding based on limit exceedances. We present two types of maps in this section. The first one is a distributed map based on the water depth levels in relation to the topographic level (Figs. 7 and 8). The second type is a graduated symbol map, where the size of the circles indicates the probability of exceedance given the depth of a specific house cellar (Fig. 9).

Figs. 7 and 8 show the probability of having groundwater tables rising at the defined thresholds  $\kappa_{ij}$  ( $d = 1.5$ ) and  $\kappa_{ij}$  ( $d = 0.5$ ), respectively. The maps include the outcomes of simulations with the prior uncertainty (Figs. 7a and 8a) and the revised uncertainty from the Bayesian inversion (Figs. 7b and 8b). Areas with a low probability of exceedance are reduced in the maps built with the posterior distributions, thanks to the reduced variability in the sensitive parameters. Many



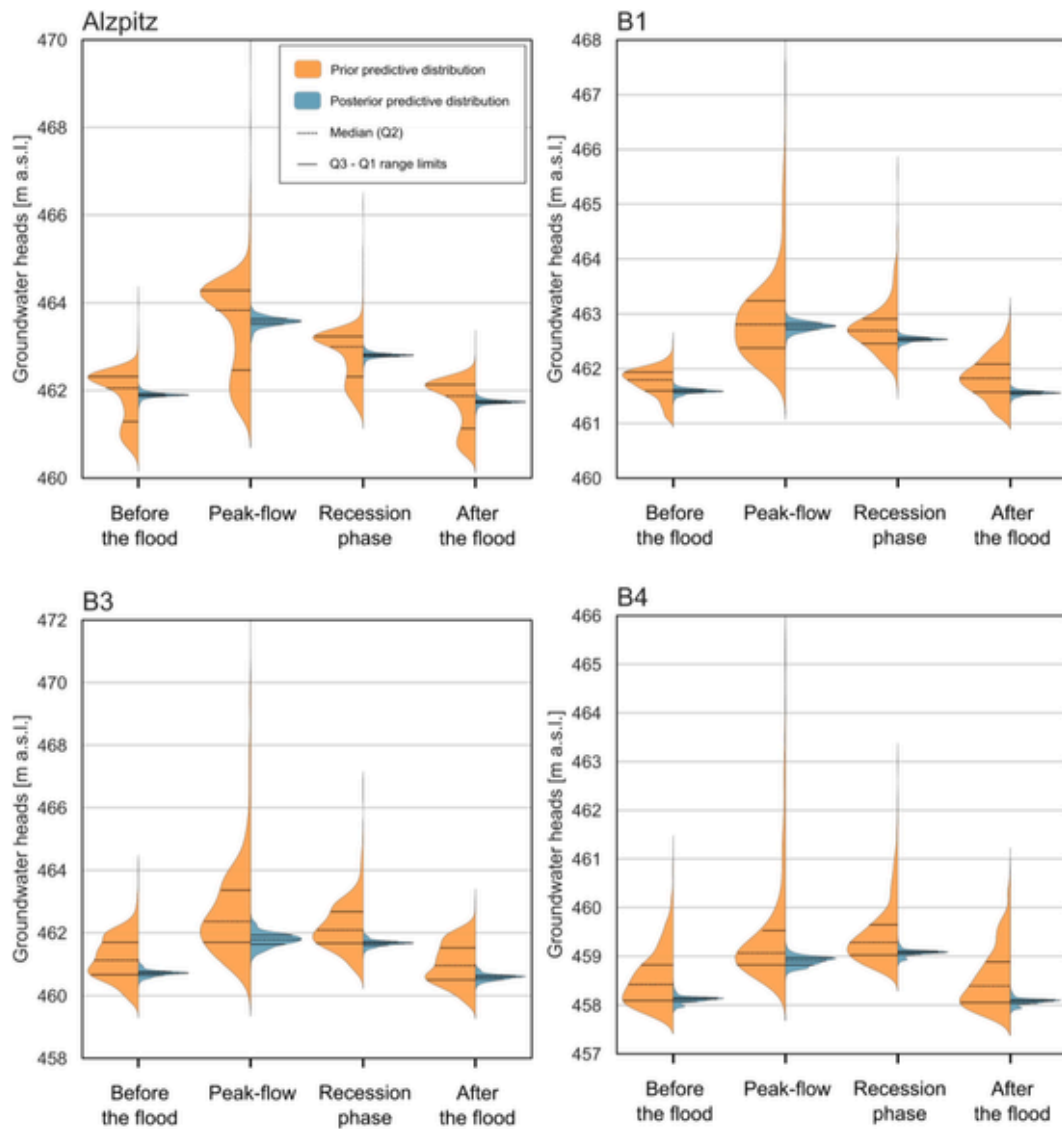
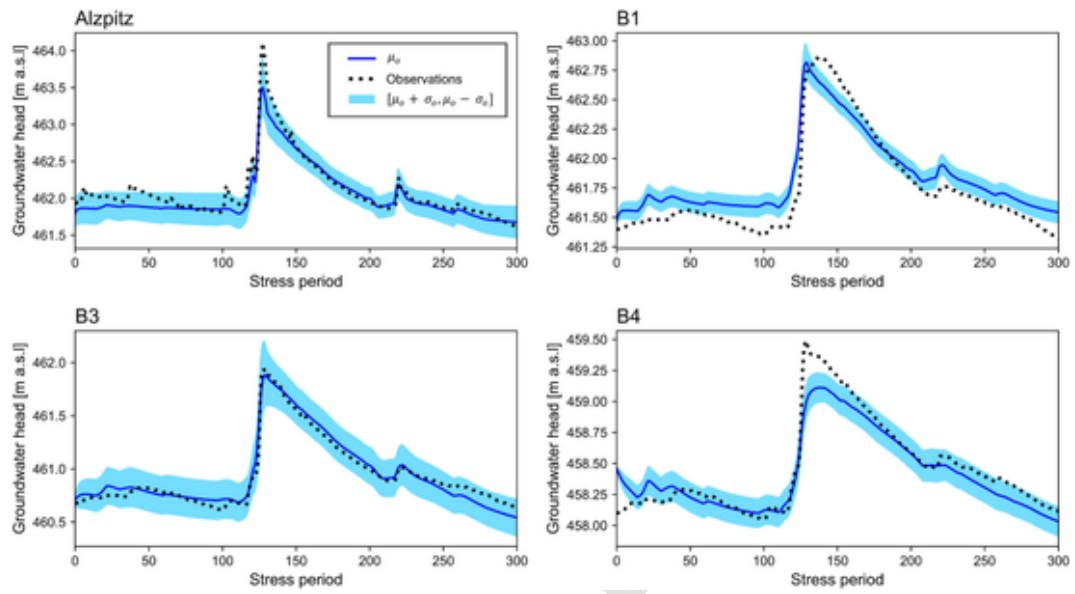


Fig. 5. Violin plot of the probability density function of the groundwater heads from the prior (orange) and the posterior (blue) predictive distributions at observation wells and at different phases of the event. The second quartile (i.e., median) of the distribution is illustrated with the dashed line. First (Q1) and third quartiles (Q3) are represented by the continuous line. Probability distributions are scaled to obtain prior and posterior violins with the same width, so that the differences in their shapes are easily perceptible.

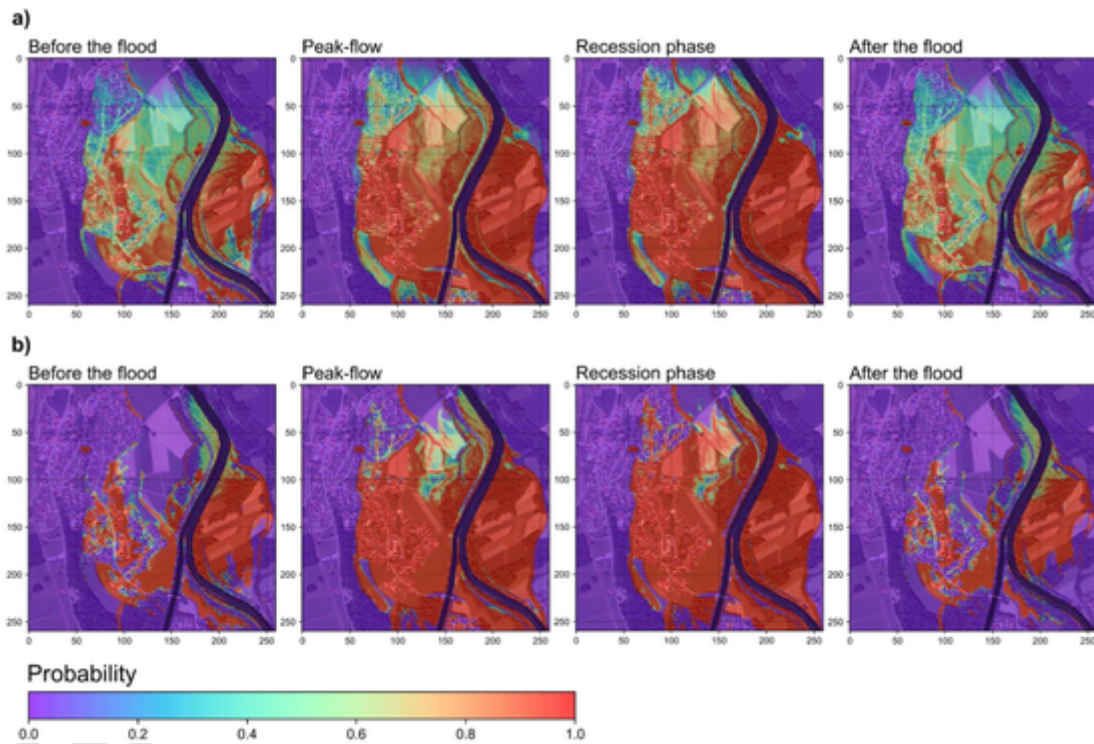
zones are highly susceptible to groundwater flooding, even before and after the peak-flow. This agrees with reports of cellar inundations out of the peak-flow event and the high groundwater levels that can also be observed during non-flood seasons (Bayerisches Landesamt für Umwelt, 2020). The heavy precipitation at the end of May and the beginning of June 2013 (between stress period 115 and stress period 130) could have been the inundation catalyst, but the results show a high groundwater table at every phase of the event regardless of the heavy rainfall. Indeed, initial conditions, such as soil saturation before the event can influence recharge from rainfall. The underground infrastructure and properties can hence be exposed during longer periods to groundwater flooding and not only during specific hydrological events. The uncertainty in the soil saturation, parametrized using the recharge multiplier in this study, is essential because the occurrence, duration and severity of groundwater flooding will be determined by both the antecedent water table and groundwater recharge rate.

During the peak-flow and the recession phase, the probability of having groundwater depths above 1.5 m reaches maximum values. This shows the relatively slow decrease in the hydraulic head in the recession phase in comparison with fluvial floods. Groundwater flooding can

last for an extended period, and infrastructure and property damage can be associated with long-lasting exposure. By comparing the outcomes from prior and posterior evaluations, we observe cells where the prior beliefs overestimate the exceedance probabilities while others underestimate them. In such cases, preconceptions based on one single forward simulation could be misleading. Maps over the basis of a distributed model help to avoid such biases and identify spatial responses that may be overlooked otherwise. Also, creating the maps with a Bayesian approach increases the spatial confidence of the groundwater flooding susceptibility. We mainly assign this improvement to the updated posterior distributions of the hydraulic conductivity ( $k_A$  and  $k_B$ ), which are very sensitive parameters and describe the ability of the material to transmit the fluid through the pore space within significant regions of the domain. Comparing Figs. 7 and 8, we can observe that the choice in the threshold value  $d$  is very important for the delineation of the areas affected by groundwater flooding. Hence, it should be carefully chosen according to site-specific problematics or different values of  $d$  should be investigated to have a clear picture of the possibly inundated areas.



**Fig. 6.** Predictive uncertainty for the groundwater flood event at different monitoring wells. the blue shade in the plots represents the interval  $[\mu_o + \sigma_o, \mu_o - \sigma_o]$ . The solid blue line indicates the mean of the posterior predictive distribution of the groundwater heads and the dotted line indicates the groundwater head observations.

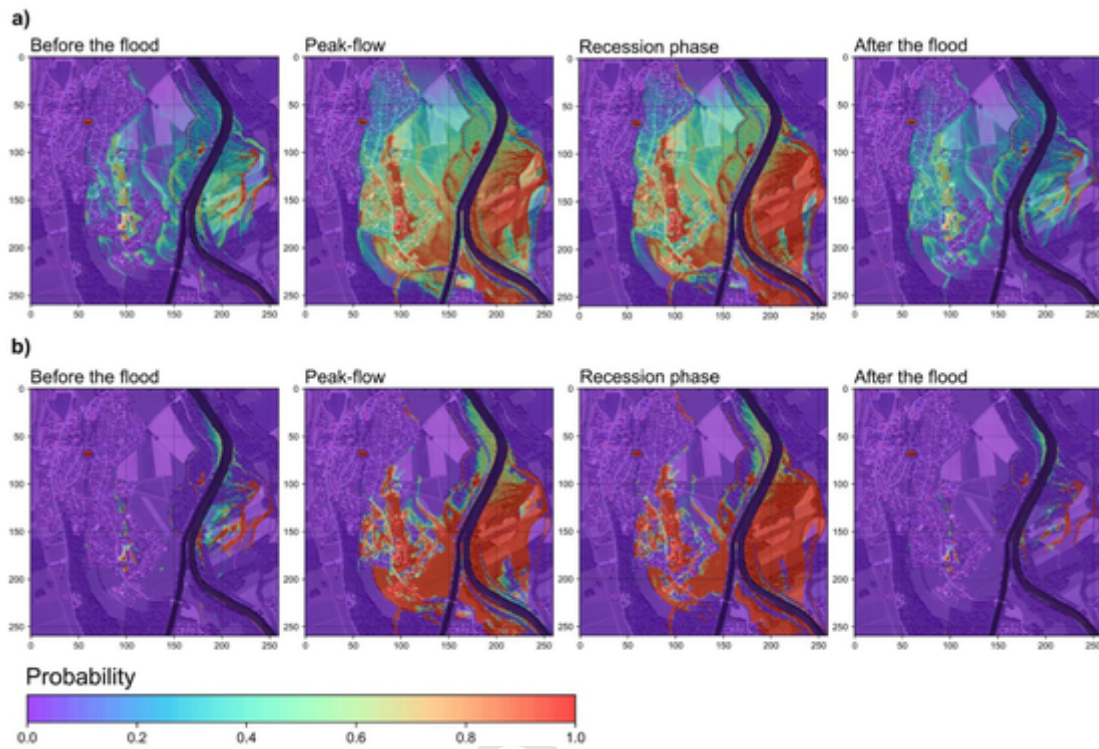


**Fig. 7.** Probability maps of groundwater depth equal to or below 1.5 m at different phases of the event. Maps are created with a) prior predictive distribution and b) posterior predictive distribution.

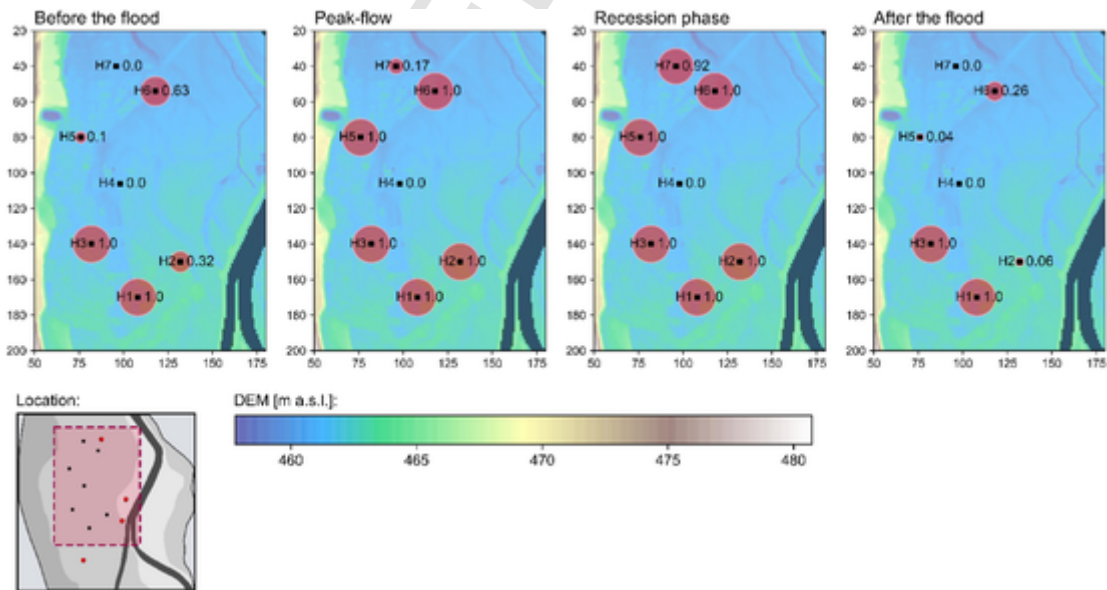
Based on the household survey, we define the threshold  $\hat{\kappa}_{i,j}$  at different residential properties and the resulting maps can be observed in Fig. 9. The properties selected in this analysis are far enough from the boundary conditions such that they are not influenced by the time-variant specified heads. We observe that the probability of reaching the cellar depth is unrelated to the distance to the river (see probabilities of building H2 and H3). Given the high groundwater table of the simulated period, it is very likely to reach the cellar depths at H1, H2, H3, and H6 even before and after the peak event. The probability of groundwater levels reaching the cellar is higher at the recession phase at H2,

H5, H6, and H7. These results may imply a delayed propagation of the flood wave from the river into the aquifer.

A drawback of the presented scheme is the memory requirement to store the element-wise data to construct the probability density functions. Still, in practice, it is possible to cluster the output data in relation to their spatial and temporal relevance and build maps of smaller regions. This means, once the zones and periods of major importance are identified, there is no need to keep in memory the simulation data that is out of the cluster. The presented results are obtained from a groundwater flow model that evaluates one actual event, which involves a rel-



**Fig. 8.** Probability maps of groundwater depth equal to or below 0.5 m at different phases of the event. Maps are created with a) prior predictive distribution and b) posterior predictive distribution.



**Fig. 9.** Probability of groundwater levels rising to or above the depths of the cellars. Graduated symbols (red circles) indicate the probability. The image displays a specific section of the model domain.

actively short timeframe of simulation, compared to mapping products that measure susceptibility to groundwater flooding from the description of the return period. However, the approach can be easily extended considering appropriate design parameters (e.g., changing the return period of the precipitation event, of the peak river discharge and/or of the volume of the event). This single event approach could be informative for industries, property managers, or insurance companies to evaluate the susceptibility to groundwater flooding of underground structures.

#### 4. Conclusions

The broad implication of this research is the presentation of a Bayesian-based framework to create probability maps of susceptibility to groundwater flooding in order to incorporate the parameter uncertainties in the risk imagery. The scheme proposes to break down the uncertainty and sensitivity of one flood event into spatial and temporal terms and compute the statistics of interest at each domain element. The elementary effects method is used to reduce the problem dimensionality by quantifying the model sensitivity with a relatively low number of model evaluations. We use the DREAM algorithm as an



MCMC-based technique to estimate the posterior distributions of the relevant uncertain parameters and reduce the uncertainty in the predicted groundwater heads. Finally, we propose to explore the posterior probability distribution of the groundwater heads to compute the exceedance probability based on the groundwater depths and cellar depths. Two types of maps are created to describe the susceptibility to groundwater flooding: a distributed map using the groundwater depth threshold, and a graduated symbol map given the depth of a specific house cellar.

The approach is illustrated with an application in a numerical model that describes the groundwater flood event in the Alz River valley in 2013. We are able to increase the spatial confidence of the results thanks to the reduction in the uncertainty during the inference process. By using a relatively large temporal scale, we are able to employ accurate temporal resolutions to detect the exposure to groundwater inundation at particular properties and areas. The framework can be applied over one flood event and this may be considered a promising aspect due to the reduced amount of data and the low computational demand. The posterior probabilities of the groundwater head and the exceedance evaluation help us distinguish highly exposed zones from those exposed only during peak-flow and the recession phase. This is relevant in practical applications because definitions of the susceptible zones can be easily constructed from a probabilistic perspective, and the accuracy of the simulations can be updated with new evidence data. In addition, these findings provide additional information about the incidence of groundwater flooding beyond the flood season. In this specific case study, the results indicate that infrastructure and property can be long-lastingly exposed to inundation hazards.

Finally, we stress the importance of opting for a Bayesian approach to understand groundwater flood events and create risk imagery. The temporal and spatial nature of groundwater flow confirms that the problem can be presented from a different perspective than the view of traditional flood events. Besides the exposure of the assets, groundwater flooding risk depends on temporal and spatial details from hydrological, climatic, and geomorphological elements. Ultimately, this level of detail makes flood mapping the crucial tool for flood risk management employed by a wide variety of partakers (such as planners, developers, insurance companies, and property owners). But it is not possible to characterize the hydrogeological systems in such detail without being affected by uncertainty, and an explicit quantification is therefore required. Probability maps built over inverse methods can be relevant not only for incorporating the uncertainty and improving the reliability of the model but also for the accuracy of the message that is transmitted to different actors.

#### CRediT authorship contribution statement

**Pablo Merchán-Rivera:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Alexandra Geist:** Validation, Formal analysis, Investigation, Writing – original draft. **Markus Disse:** Conceptualization, Writing – review & editing, Supervision. **Jingshui Huang:** Conceptualization, Writing – review & editing. **Gabriele Chiogna:** Conceptualization, Resources, Methodology, Investigation, Writing – review & editing, Supervision, Project administration.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgements

The first author is supported by grant CZO2-11621 of the Secretaría Nacional de Educación, Ciencia y Tecnología de Ecuador (SENESCYT). This research is part of the UNMIX Project (Uncertainties due to boundary conditions in predicting mixing in groundwater), which is supported by the Deutsche Forschungsgemeinschaft (DFG), GSC 81, through the TUM International Graduate School for Science and Engineering (IGSSE). The dataset, code and the simulation results can be found in the repository (Merchán-Rivera, 2021), with license CC BY 4.0. Finally, we want to acknowledge to the authors and contributors of the various Python libraries used in this work for their enormous contribution to science and education. We also thank the reviewers for their valuable comments.

#### References

- Abdoud, J.M., Ryan, M.C., Osborn, G.D., 2018. Groundwater flooding in a river-connected alluvial aquifer. *J. Flood Risk Manag.* 11, e12334. <https://doi.org/10.1111/jfr3.12334>.
- Allocca, V., Di Napoli, M., Coda, S., Carotenuto, F., Calcaterra, D., Di Martire, D., De Vita, P., 2021. A novel methodology for Groundwater Flooding Susceptibility assessment through Machine Learning techniques in a mixed-land use aquifer. *Sci. Total Environ.* 790, 148067. <https://doi.org/10.1016/j.scitotenv.2021.148067>.
- Ascott, M.J., Marchant, B.P., Macdonald, D., McKenzie, A.A., Bloomfield, J.P., 2017. Improved understanding of spatio-temporal controls on regional scale groundwater flooding using hydrograph analysis and impulse response functions. *Hydrol. Process.* 31, 4586–4599. <https://doi.org/10.1002/hyp.11380>.
- Bayerisches Landesamt für Umwelt, 2020. Haupttabelle für Grundwasserstände. Bayerisches Landesamt für Umwelt, 2017. UmweltAtlas Bayern.
- Beven, K., Freer, J., 2001. Equifinality, data assimilation, and uncertainty estimation in mechanistic modelling of complex environmental systems using the GLUE methodology. *J. Hydrol.* 249, 11–29. [https://doi.org/10.1016/S0022-1694\(01\)00421-8](https://doi.org/10.1016/S0022-1694(01)00421-8).
- Bhola, P.K., Leandro, J., Disse, M., 2020. Building hazard maps with differentiated risk perception for flood impact assessment. *Nat. Hazards Earth Syst. Sci.* 20, 2647–2663. <https://doi.org/10.5194/nhess-20-2647-2020>.
- Bittner, D., Engel, M., Wohlmuth, B., Labat, D., Chiogna, G., 2021. Temporal Scale-Dependent Sensitivity Analysis for Hydrological Model Parameters Using the Discrete Wavelet Transform and Active Subspaces. *Water Res.* 57. <https://doi.org/10.1029/2020WR028511>.
- Booth, E.G., Zipper, S.C., Loheide, S.P., Kucharik, C.J., 2016. Is groundwater recharge always serving us well? Water supply provisioning, crop production, and flood attenuation in conflict in Wisconsin, USA. *Ecosyst. Serv.* 21, 153–165. <https://doi.org/10.1016/j.ecoser.2016.08.007>.
- Box, G.E.P., Tiao, G.C., 2011. *Bayesian Inference in Statistical Analysis*. John Wiley & Sons.
- Boyce, S.E., Yeh, W.-W.-G., 2014. Parameter-independent model reduction of transient groundwater flow models: Application to inverse problems. *Adv. Water Resour.* 69, 168–180. <https://doi.org/10.1016/j.advwatres.2014.04.009>.
- British Geological Survey, 2006. Groundwater Flooding Susceptibility (Metadata Catalogue). British Geological Survey, United Kingdom.
- Campolongo, F., Cariboni, J., Saltelli, A., 2007. An effective screening design for sensitivity analysis of large models. *Environmental Modelling & Software*, Modelling, computer-assisted simulations, and mapping of dangerous phenomena for hazard assessment 22, 1509–1518. <https://doi.org/10.1016/j.envsoft.2006.10.004>.
- Collins, S.L., Christelis, V., Jackson, C.R., Mansour, M.M., Macdonald, D.M.J., Barkwith, A.K.A.P., 2020. Towards integrated flood inundation modelling in groundwater-dominated catchments. *J. Hydrol.* 591, 125755. <https://doi.org/10.1016/j.jhydrol.2020.125755>.
- Colombo, L., Gattinoni, P., Scesi, L., 2018. Stochastic modelling of groundwater flow for hazard assessment along the underground infrastructures in Milan (northern Italy). *Tunn. Undergr. Space Technol.* 79, 110–120. <https://doi.org/10.1016/j.tust.2018.05.007>.
- Demeritt, D., Nobert, S., 2014. Models of best practice in flood risk communication and management. *Environ. Hazards* 13, 313–328. <https://doi.org/10.1080/17477891.2014.924897>.
- DHI, 2013. The MIKE SHE user and technical reference manual. Danish Hydraulic Institute, Copenhagen.
- Di Baldassarre, G., Schumann, G., Bates, P.D., Freer, J.E., Beven, K.J., 2010. Flood-plain mapping: a critical discussion of deterministic and probabilistic approaches. *Hydrol. Sci. J.* 55, 364–376. <https://doi.org/10.1080/02626661003683389>.
- Disse, M., Keilholz, P., Bhola, P., 2015. Die Grundhochwasser - Problematik in der Gemeinde Tacherting und Untersuchung von Gegenmaßnahmen. Technical University of Munich, Munich, Chair of Hydrology and River Basin Management.
- Doppler, G., Kroemer, E., Rögner, K., Wallner, J., Jerz, H., Grotenthaler, W., 2011. Quaternary Stratigraphy of Southern Bavaria. <https://doi.org/10.23689/FIDGEO-1763>.
- Erdal, D., Cirpka, O.A., 2019. Global sensitivity analysis and adaptive stochastic sampling of a subsurface-flow model using active subspaces. *Hydrol. Earth Syst. Sci.* 23, 3787–3805. <https://doi.org/10.5194/hess-23-3787-2019>.



- Feng, K., Lu, Z., Yang, C., 2019. Enhanced Morris method for global sensitivity analysis: good proxy of Sobol' index. *Struct. Multidisc. Optim.* 59, 373–387. <https://doi.org/10.1007/s00158-018-2071-7>.
- Fetter, C.W., 1999. *Contaminant hydrogeology*, 2nd ed. Prentice Hall, Upper Saddle River, NJ.
- Fürst, J., Bichler, A., Konecny, F., 2015. Regional Frequency Analysis of Extreme Groundwater Levels. *Groundwater* 53, 414–423. <https://doi.org/10.1111/gwat.12223>.
- García Sanchez, D., Lacarrière, B., Musy, M., Bourges, B., 2014. Application of sensitivity analysis in building energy simulations: Combining first- and second-order elementary effects methods. *Energy Build.* 68, 741–750. <https://doi.org/10.1016/j.enbuild.2012.08.048>.
- Gattinoni, P., Scesi, L., 2017. The groundwater rise in the urban area of Milan (Italy) and its interactions with underground structures and infrastructures. *Tunn. Undergr. Space Technol.* 62, 103–114. <https://doi.org/10.1016/j.tust.2016.12.001>.
- Gelman, A., Rubin, D.B., 1992. Inference from Iterative Simulation Using Multiple Sequences. *Statist. Sci.* 7. <https://doi.org/10.1214/ss/1177011136>.
- González-Quirós, A., Fernández-Álvarez, J.P., 2019. Conceptualization and finite element groundwater flow modeling of a flooded underground mine reservoir in the Asturian Coal Basin, Spain. *J. Hydrol.* 578, 124036. <https://doi.org/10.1016/j.jhydrol.2019.124036>.
- Guillaume, J.H.A., Hunt, R.J., Comunian, A., Blakers, R.S., Fu, B., 2016. Methods for Exploring Uncertainty in Groundwater Management Predictions, in: Jakeman, A.J., Barreteau, O., Hunt, R.J., Rinaudo, J.-D., Ross, A. (Eds.), *Integrated Groundwater Management*. Springer International Publishing, Cham, pp. 711–737. [https://doi.org/10.1007/978-3-319-23576-9\\_28](https://doi.org/10.1007/978-3-319-23576-9_28).
- Harbaugh, A., 2005. MODFLOW-2005, The U.S. Geological Survey Modular Ground-Water Model—the Ground-Water Flow Process, U.S. Geological Survey Techniques and Methods 6-A16. U.S. Geological Survey.
- Harbaugh, A.W., Banta, E.R., Hill, M.C., McDonald, M.G., 2000. MODFLOW-2000, The U.S. Geological Survey Modular Ground-Water Model - User Guide to Modularization Concepts and the Ground-Water Flow Process (USGS Numbered Series No. 2000-92), MODFLOW-2000, The U.S. Geological Survey Modular Ground-Water Model - User Guide to Modularization Concepts and the Ground-Water Flow Process, Open-File Report. Geological Survey (U.S.). <https://doi.org/10.3133/ofr200092>.
- Harris, C.R., Millman, K.J., van der Walt, S.J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N.J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M.H., Brett, M., Haldane, A., del Río, J.F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., Oliphant, T.E., 2020. Array programming with NumPy. *Nature* 585, 357–362. <https://doi.org/10.1038/s41586-020-2649-2>.
- Helton, J.C., Johnson, J.D., Oberkampf, W.L., 2004. An exploration of alternative approaches to the representation of uncertainty in model predictions. *Reliability Engineering & System Safety*, Alternative Representations of Epistemic Uncertainty 85, 39–71. <https://doi.org/10.1016/j.res.2004.03.025>.
- Henstra, D., Minano, A., Thistlethwaite, J., 2019. Communicating disaster risk? An evaluation of the availability and quality of flood maps. *Nat. Hazards Earth Syst. Sci.* 19, 313–323. <https://doi.org/10.5194/nhess-19-313-2019>.
- Herman, J., Usher, W., 2017. SALib: An open-source Python library for Sensitivity Analysis. *J. Open Source Softw.* 2, 97. <https://doi.org/10.21105/joss.00097>.
- Heße, F., Comunian, A., Attinger, S., 2019. What We Talk About When We Talk About Uncertainty. Toward a Unified, Data-Driven Framework for Uncertainty Characterization in Hydrogeology. *Front. Earth Sci.* 7, 118. <https://doi.org/10.3389/feart.2019.00118>.
- Houska, T., Kraft, P., Chamorro-Chavez, A., Breuer, L., 2015. SPOTting Model Parameters Using a Ready-Made Python Package. *PLoS ONE* 10, e0145180. <https://doi.org/10.1371/journal.pone.0145180>.
- Hunter, J.D., 2007. Matplotlib: A 2D Graphics Environment. *Comput. Sci. Eng.* 9, 90–95. <https://doi.org/10.1109/MCSE.2007.55>.
- Jaxa-Rozen, M., Kwakkel, J., 2018. Tree-based ensemble methods for sensitivity analysis of environmental models: A performance comparison with Sobol and Morris techniques. *Environ. Modell. Software* 107, 245–266. <https://doi.org/10.1016/j.envsoft.2018.06.011>.
- Jimenez-Martinez, J., Smith, M., Pope, D., 2015. Prediction of groundwater-induced flooding in a chalk aquifer for future climate change scenarios: Prediction of Groundwater-Induced Flooding in a Chalk Aquifer. *Hydrol. Process.* <https://doi.org/10.1002/hyp.10619>.
- Kaipio, J., Somersalo, E., 2005. *Statistical and Computational Inverse Problems*, Applied Mathematical Sciences. Springer-Verlag, New York. <https://doi.org/10.1007/b138659>.
- Kaiser, M., Borga, M., Disse, M., 2020. Occurrence and Characteristics of Flash Floods in Bavaria (Germany). In: Leal Filho, W., Nagy, G.J., Borga, M., Chávez Muñoz, P.D., Magnuszewski, A. (Eds.), *Climate Change, Hazards and Adaptation Options*, Climate Change Management. Springer International Publishing, Cham, pp. 293–310. [https://doi.org/10.1007/978-3-030-37425-9\\_16](https://doi.org/10.1007/978-3-030-37425-9_16).
- Keilholz, P., Disse, M., Bhola, P., 2015. Integrierte Betrachtung der Grundhochwasser-Problematik in der Gemeinde Tacherting (Bayern). *Hydrol. Wasserbewirtschaft.* 2015, 688–693. <https://doi.org/10.3243/kwe2015.11.003>.
- Kreibich, H., Thieken, A.H., Grunenberg, H., Ullrich, K., Sommer, T., 2009. Extent, perception and mitigation of damage due to high groundwater levels in the city of Dresden, Germany. *Nat. Hazards Earth Syst. Sci.* 9, 1247–1258. <https://doi.org/10.5194/nhess-9-1247-2009>.
- Leake, S.A., Prudic, D.E., 1991. Documentation of a computer program to simulate aquifer-system compaction using the modular finite-difference ground-water flow model (No. 06-A2), Techniques of Water-Resources Investigations. U.S. Geological Survey.
- Macdonald, D., Bloomfield, J.P., Hughes, A.G., MacDonald, A., Adams, B., McKenzie, A.A., 2008. Improving the understanding of the risk from groundwater flooding in the UK. Presented at the FLOODrisk 2008, European Conference on Flood Risk Management, CRC Press, The Netherlands.
- Macdonald, D., Dixon, A., Newell, A., Hallaways, A., 2012. Groundwater flooding within an urbanised flood plain: Groundwater flooding within urbanised flood plain. *J. Flood Risk Manage.* 5, 68–80. <https://doi.org/10.1111/j.1753-318X.2011.01127.x>.
- Mancini, C.P., Lollai, S., Volpi, E., Fiori, A., 2020. Flood Modeling and Groundwater Flooding in Urbanized Reclamation Areas: The Case of Rome (Italy). *Water* 12, 2030. <https://doi.org/10.3390/w12072030>.
- Mara, T.A., Fajraoui, N., Guadagnini, A., Younes, A., 2017. Dimensionality reduction for efficient Bayesian estimation of groundwater flow in strongly heterogeneous aquifers. *Stoch. Environ. Res. Risk Assess.* 31, 2313–2326. <https://doi.org/10.1007/s00477-016-1344-1>.
- McKenzie, A.A., Rutter, H.K., Hulbert, A.G., 2010. The use of elevation models to predict areas at risk of groundwater flooding. *Geol. Soc. Lond. Spec. Publ.* 345, 75–79. <https://doi.org/10.1144/SP345.9>.
- Merchán-Rivera, P., 2021. Dataset and algorithms for the Bayesian framework to assess and create risk maps of groundwater flooding. Mendeley Data. <https://doi.org/10.17632/2j8hy7jfr2.2>.
- Merchán-Rivera, P., Chiogna, G., Disse, M., Bhola, P., 2018. Surface water and groundwater interaction during flood events in the Alz Valley: Numerical modeling and solute transport simulations, in: *Agua Subterránea., Interacción de Agua Superficial y Agua Subterránea*. Editorial de la Universidad Nacional de Salta, Salta, Argentina.
- Merchán-Rivera, P., Wohlmuth, B., Chiogna, G., 2021. Identifying Stagnation Zones and Reverse Flow Caused by River-Aquifer Interaction: An Approach Based on Polynomial Chaos Expansions. *Water Res.* <https://doi.org/10.1029/2021WR029824>.
- Merz, B., Thieken, A.H., 2009. Flood risk curves and uncertainty bounds. *Nat. Hazards* 51, 437–458. <https://doi.org/10.1007/s11069-009-9452-6>.
- Morris, M.D., 1991. Factorial Sampling Plans for Preliminary Computational Experiments. *Technometrics* 33, 161–174. <https://doi.org/10.1080/00401706.1991.10484804>.
- Morris, S.E., Cobby, D., Zaidman, M., Fisher, K., 2018. Modelling and mapping groundwater flooding at the ground surface in Chalk catchments: Modelling and mapping groundwater flooding. *J. Flood Risk Manage.* 11, S251–S268. <https://doi.org/10.1111/jfr3.12201>.
- Morrissey, P., McCormack, T., Naughton, O., Meredith Johnston, P., William Gill, L., 2020. Modelling groundwater flooding in a lowland karst catchment. *J. Hydrol.* 580, 124361. <https://doi.org/10.1016/j.jhydrol.2019.124361>.
- Mustafa, S.M.T., Nossent, J., Ghysels, G., Huysmans, M., 2018. Estimation and Impact Assessment of Input and Parameter Uncertainty in Predicting Groundwater Flow With a Fully Distributed Model. *Water Resour. Res.* 54, 6585–6608. <https://doi.org/10.1029/2017WR021857>.
- Oyedele, K.F., Ayolabi, E.A., Adeoti, L., Adegbola, R.B., 2009. Geophysical and hydrogeological evaluation of rising groundwater level in the coastal areas of Lagos, Nigeria. *Bull. Eng. Geol. Environ.* 68, 137–143. <https://doi.org/10.1007/s10064-008-0182-x>.
- Preen, M., Fisher, S., 2015. Impacts from groundwater control in urban areas 6.
- Ross, J.L., Ozbek, M.M., Pinder, G.F., 2009. Aleatoric and epistemic uncertainty in groundwater flow and transport simulation. *Water Resour. Res.* 45. <https://doi.org/10.1029/2007WR006799>.
- Saltelli, A., 2002. Sensitivity Analysis for Importance Assessment. *Risk Anal.* 22, 579–590. <https://doi.org/10.1111/0272-4332.00040>.
- Saltelli, A., Tarantola, S., Campolongo, F., Ratto, M., 2004. *Sensitivity Analysis in Practice: A Guide to Assessing Scientific Models*. John Wiley & Sons.
- Scheibe, T.D., Murphy, E.M., Chen, X., Rice, A.K., Carroll, K.C., Palmer, B.J., Tartakovsky, A.M., Battiatto, L., Wood, B.D., 2015. An Analysis Platform for Multiscale Hydrogeologic Modeling with Emphasis on Hybrid Multiscale Methods. *Groundwater* 53, 38–56. <https://doi.org/10.1111/gwat.12179>.
- Schoups, G., Vrugt, J.A., 2010. A formal likelihood function for parameter and predictive inference of hydrologic models with correlated, heteroscedastic, and non-Gaussian errors 2009WR008933. *Water Resour. Res.* 46. <https://doi.org/10.1029/2009WR008933>.
- Smith, R.C., 2013. *Uncertainty Quantification: Theory, Implementation, and Applications*. SIAM, Computational Science and Engineering.
- Soren, J., 1976. Basement flooding and foundation damage from water-table rise in the East New York section of Brooklyn, Long Island, New York (USGS Numbered Series No. 76-95), Basement flooding and foundation damage from water-table rise in the East New York section of Brooklyn, Long Island, New York, Water-Resources Investigations Report. U.S. Geological Survey. <https://doi.org/10.3133/wri7695>.
- Stanko, Z.P., Boyce, S.E., Yeh, W.-W.-G., 2016. Nonlinear model reduction of unconfined groundwater flow using POD and DEIM. *Adv. Water Resour.* 97, 130–143. <https://doi.org/10.1016/j.advwatres.2016.09.005>.
- Tabari, H., 2020. Climate change impact on flood and extreme precipitation increases with water availability. *Sci. Rep.* 10, 13768. <https://doi.org/10.1038/s41598-020-70816-2>.
- Tarantola, A., 2005. Inverse Problem Theory and Methods for Model Parameter Estimation. SIAM.
- Teixeira Parente, M., Bittner, D., Mattis, S.A., Chiogna, G., Wohlmuth, B., 2019. Bayesian Calibration and Sensitivity Analysis for a Karst Aquifer Model Using Active Subspaces. *Water Resour. Res.* 55, 7086–7107. <https://doi.org/10.1029/2019WR024739>.
- Teng, J., Jakeman, A.J., Vaze, J., Croke, B.F.W., Dutta, D., Kim, S., 2017. Flood inundation modelling: A review of methods, recent advances and uncertainty analysis. *Environ. Modell. Software* 90, 201–216. <https://doi.org/10.1016/j.envsoft.2017.01.006>.
- Upton, K.A., Jackson, C.R., 2011. Simulation of the spatio-temporal extent of groundwater flooding using statistical methods of hydrograph classification and lumped parameter

- models. *Hydrol. Process.* 25, 1949–1963. <https://doi.org/10.1002/hyp.7951>.
- Vrugt, J.A., 2016. Markov chain Monte Carlo simulation using the DREAM software package: Theory, concepts, and MATLAB implementation. *Environ. Modell. Software* 75, 273–316. <https://doi.org/10.1016/j.envsoft.2015.08.013>.
- Vrugt, J.A., Ter Braak, C.J.F., 2011. DREAM(D): an adaptive Markov Chain Monte Carlo simulation algorithm to solve discrete, noncontinuous, and combinatorial posterior parameter estimation problems. *Hydrol. Earth Syst. Sci.* 15, 3701–3713. <https://doi.org/10.5194/hess-15-3701-2011>.
- Vrugt, J.A., ter Braak, C.J.F., Clark, M.P., Hyman, J.M., Robinson, B.A., 2008. Treatment of input uncertainty in hydrologic modeling: Doing hydrology backward with Markov chain Monte Carlo simulation. *Water Resour. Res.* 44. <https://doi.org/10.1029/2007WR006720>.
- Vrugt, J.A., ter Braak, C.J.F., Diks, C.G.H., Robinson, B.A., Hyman, J.M., Higdon, D., 2009. Accelerating Markov Chain Monte Carlo Simulation by Differential Evolution with Self-Adaptive Randomized Subspace Sampling. *Int. J. Nonlin. Sci. Numer. Simul.* 10. <https://doi.org/10.1515/IJNSNS.2009.10.3.273>.
- Willems, W., 2011. Unsicherheit von Wasserstands- und Abflussmessungen an A-Pegeln des Landes Bayern. Auftragsstudie für das Bayerische Landesamt für Umwelt, LfU Bayern.
- Willems, W., Stricker, K., 2012. Bewertung der Qualität von Abflusszeitreihendaten anhand eines mathematisch/statistisch ermittelten Unsicherheitskriteriums. Auftragsstudie für das Bayerische Landesamt für Umwelt, LfU Bayern.
- Winsemius, H.C., Aerts, J.C.J.H., van Beek, L.P.H., Bierkens, M.F.P., Bouwman, A., Jongman, B., Kwadijk, J.C.J., Ligtoet, W., Lucas, P.L., van Vuuren, D.P., Ward, P.J., 2016. Global drivers of future river flood risk. *Nat. Clim. Change* 6, 381–385. <https://doi.org/10.1038/nclimate2893>.
- Xu, T., Valocchi, A.J., Ye, M., Liang, F., Lin, Y., 2017. Bayesian calibration of groundwater models with input data uncertainty. *Water Resour. Res.* 53, 3224–3245. <https://doi.org/10.1002/2016WR019512>.
- Yu, X., Moraetis, D., Nikolaidis, N.P., Li, B., Duffy, C., Liu, B., 2019. A coupled surface-subsurface hydrologic model to assess groundwater flood risk spatially and temporally. *Environ. Modell. Software* 114, 129–139. <https://doi.org/10.1016/j.envsoft.2019.01.008>.
- Zhou, H., Gómez-Hernández, J.J., Li, L., 2014. Inverse methods in hydrogeology: Evolution and recent trends. *Adv. Water Resour.* 63, 22–37. <https://doi.org/10.1016/j.advwatres.2013.10.014>.